npj Digital Medicine

https://doi.org/10.1038/s41746-026-02933-7

Article in Press

Benchmarking and developing large language models using one million clinical trials

Received: 27 January 2026

Accepted: 17 June 2026

Published online: 31 July 2026

Cite this article as: Wang, Z., Lin, J., Jin, Q. *et al.* Benchmarking and developing large language models using one million clinical trials. *npj Digit. Med.* (2026). https://doi.org/10.1038/s41746-026-02933-7

Zifeng Wang, Jiacheng Lin, Qiao Jin, Junyi Gao, Jathurshan Pradeepkumar, Pengcheng Jiang, Zhiyong Lu & Jimeng Sun

Benchmarking and Developing Large Language Models Using One Million Clinical Trials

Zifeng Wang[1+,#], Jiacheng Lin[2+], Qiao Jin[3+], Junyi Gao[4,5], Jathurshan Pradeepkumar[2], Pengcheng Jiang[2], Zhiyong Lu[3], and Jimeng Sun[1,2#]

[1]*Keiji AI, Seattle, USA*

[2]*School of Computing and Data Science, University of Illinois Urbana-Champaign, Urbana, IL, USA*

[3]*Division of Intramural Research, National Library of Medicine, National Institutes of Health, Bethesda, MD, USA*

[4]*Usher Institute, Edinburgh Medical School, University of Edinburgh, Edinburgh, UK*

[5]*Health Data Research UK, UK*

[+]Equal contribution

[#]Correspondence: zifeng@keiji.ai, jimeng@illinois.edu

Project website: `https://keiji.ai/research/trialpanorama`

Abstract

Developing artificial intelligence (AI) for clinical research requires a comprehensive data foundation for model benchmarking and development. Here, we introduce `TrialPanorama`, a large-scale structured resource aggregating 1.6M clinical trial records from fifteen global registries linked with biomedical ontologies and literature. Using this resource, we construct 152K training and testing samples spanning eight clinical research tasks, including systematic review, trial design, and trial optimization. Benchmarking cutting-edge large language models (LLMs) reveals limited clinical reasoning capability in generic LLMs. In contrast, an 8B LLM developed on `TrialPanorama` using supervised finetuning and reinforcement learning outperforms 70B generic counterparts across all eight tasks, with relative improvements of 73.7%, 67.6%, 38.4%, 37.8%, 26.5%, 20.7%, 20.0%, 18.1%, and 5.2%, respectively. These results demonstrate the potential of

domain-adapted AI to improve evidence synthesis and clinical trial design, establishing `TrialPanorama` as a foundation for scaling AI in clinical research.

Introduction

Developing new drugs remains a lengthy and expensive process that often spans close to a decade and costs more than two billion dollars [1]. A major share of this burden arises during the human study phase when clinical trials are conducted to evaluate safety and efficacy across phases I through IV. Beyond the execution of trials, clinicians, researchers, and trial designers routinely search for and review clinical trial protocols and results. These activities support essential objectives such as planning new studies, assessing treatments, and informing clinical decision making in evidence-based medicine [2, 3].

There is growing interest in using artificial intelligence (AI), and especially large language models (LLMs), to improve the development and evaluation of clinical trials. Recent advances have explored a diverse set of tasks including predicting clinical trial outcomes [4–6], matching patients to appropriate trials [7, 8], extracting real world cohort data from eligibility criteria [9, 10], drafting clinical trial documents [11, 12], and accelerating systematic review of clinical studies [13, 14]. Nonetheless, most depend on specialized data sources assembled from various specialized repositories of clinical trials. As a result, progress is constrained by the absence of standardized and comprehensive clinical trial resources suitable for AI development.

In this work, we introduce `TrialPanorama`, a clinical trial database and benchmark suite created from over one million records and fifteen global sources, to provide a broad foundation for trial design, trial review, and the development of AI for clinical trial tasks. We assemble clinical trial records from major global registries, including ClinicalTrials.gov [15], the International Clinical Trials Registry Platform [16], the European Clinical Trials Register [17], and several national registries [18, 19]. We further incorporate trial-related evidence from literature sources such as PubMed. Prior studies have curated selected portions of these resources for focused objectives such as extracting trial results [20, 21], annotating drugs and conditions [22, 23], linking publications to registered trials [24], and generating structured outcome data [25]. Here, we put them into a unified representation that integrates these diverse elements, which can substantially strengthen evidence synthesis in systematic reviews and support data-driven approaches to future trial planning [26–28].

Specifically, `TrialPanorama` is created as both a database and a tool suite for extracting clinical trial information and producing benchmark data for AI development in clinical trial applications. To illustrate its utility, we present a pipeline that extracts training and testing data from the resource and supports eight trial-related tasks. Five tasks focus on trial design and optimization, including arm design, eligibility criteria design, endpoint design, sample size estimation, and trial completion assessment. Three tasks support systematic literature review workflows, including study search, study screening, and evidence question answering. We further evaluate open source and proprietary LLMs on these tasks to establish baseline performance and to demonstrate the value of `TrialPanorama` for future research.

Our benchmark further demonstrates the limitations of generic LLMs when applied to clinical trial tasks. We find that widely used LLMs display substantial gaps in study search, study screening, evidence summarization, and several trial design and optimization tasks. To explore the value of domain-specific data, we fine-tune two open-source models, Llama-3 [29] and Qwen-3 [30], using training data extracted from `TrialPanorama`. These tuned models achieve competitive or superior performance relative to larger generic LLMs across both systematic review and trial design tasks. This improvement highlights the importance of high-quality trial-specific data foundations and demonstrates that `TrialPanorama` can meaningfully strengthen the capabilities of LLMs developed for clinical trial applications.

Results

`TrialPanorama`: a unified database and knowledge graph of clinical trials

Public clinical trial information is primarily available through research publications and registry websites. To construct `TrialPanorama`, we aggregate records from PubMed and major global trial registries, including ClinicalTrials.gov and international sources such as CHICTR. In total, this yields over 1.6 million clinical trial records. As shown in Figure 1b, the final corpus consists of 48.7% PubMed-derived studies, 31.6% from ClinicalTrials.gov, and 19.6% from additional global registries. We additionally collect systematic literature reviews from PubMed and parse their cited references to identify included and excluded studies.

To harmonize these heterogeneous data sources, we process publication and registry records into a unified structured representation of trial attributes, including drugs, conditions, trial phase, endpoints, biomarkers, and outcomes. Key biomedical entities are standardized using established ontologies and databases such as DrugBank [31], MedDRA [32], RxNorm [33], MeSH [34], and COMET [35]. This processing pipeline combines rule-based methods with LLM-assisted extraction.

Using the standardized trial attributes, we construct `TrialPanorama` as both a knowledge graph and a relational database (Figure 1a). The knowledge graph links trials through shared drugs, conditions, biomarkers, and systematic review relationships, enabling structured modeling of trial relatedness. The relational database organizes detailed trial information into standardized tables covering studies, endpoints, conditions, biomarkers, adverse events, outcomes, and results. The details of database building process are in Supplementary Note 1.

As shown in Table 1, `TrialPanorama` provides broader source coverage and richer field representation than prior clinical trial databases. Figures 1c and 1d illustrate the distribution of drugs and conditions in the dataset, highlighting both frequently studied therapeutic areas and a broad long-tail of specialized trials. Overall, `TrialPanorama` serves as a comprehensive and scalable foundation for clinical trial analytics and LLM development in clinical research. We further conducted a manual quality assessment on a randomly sampled subset of the data. The results indicate that the LLM-based extraction achieves over 90% accuracy overall. Detailed results are provided in Supplementary Note 2.

A pipeline for data extraction and LLM development

`TrialPanorama` enables efficient extraction of development data for LLMs. Figure 2a illustrates an example workflow for constructing development data for the sample size estimation task. In Step 1, we identify the study fields that serve as inputs to the task. For this example, we use study-level attributes such as study type, year, phase, condition, and related descriptors as the trial setup, and use the field target accrual as the sample size value to be predicted. We further select the studies with full text so as to extract the statistical assumptions stated in the statistical analysis plan section. In Step 2, we organize these inputs into a standardized format for model development, which includes the instruction that specifies the task, the input that provides the trial setup as context, and the answer that contains the ground truth value for supervised learning.

As shown in Figures 2b and 2d, we use the `TrialPanorama` pipeline to extract development data for three systematic review tasks and five clinical trial design tasks. In systematic review task suites, the study search task takes a review topic as input and identifies the most relevant clinical trials from literature sources such as PubMed. The study screening task takes the review selection criteria and a candidate study set and evaluates the eligibility of each study for inclusion. The evidence summarization task takes the results of multiple clinical trials and produces a concise evidence summary that can support future clinical decision-making. For clinical trial design tasks, we include four tasks focused on study design fields: sample size estimation, criteria design, endpoint design, and arm design. We also include a trial completion assessment task that takes the trial setup as input, predicts whether the trial is likely to complete or terminate, and explains the factors that drive the prediction. We hold out the latest studies as the test set to avoid the data leakage issues. Notably, although the database is constructed using extensive LLM-based extraction, the ground-truth labels used in these benchmarks are directly derived from reference trial fields, reflecting expert-designed annotations from historical trials and thus not affected by LLM extraction errors. The only exception is the termination type classification, which is generated by an LLM; for this component, we report manual assessment results of label quality in Supplementary Note 2.

In Figure 2a, Step 3 applies a combination of supervised fine-tuning (SFT) and reinforcement learning with verifiable reward (RLVR) to adapt generic large language models to clinical research tasks. We first train models with SFT for all tasks, then apply RLVR to tasks that require deeper reasoning and offer a verifiable reward signal, such as sample size estimation. This staged approach enables broad task coverage while strengthening reasoning ability. Building on this framework, `TrialPanorama` benchmark provides a unified testbed to evaluate both open and proprietary models and to develop domain-adapted variants. For example, we adapt Qwen3-8B to clinical research by training it on the `TrialPanorama` corpus, producing Qwen3-8B-TP. As shown in Figure 2c, Qwen3-8B-TP delivers clear gains over the base Qwen3-8B across all tasks. Remarkably, this 8B vertical model surpasses Llama-70B, despite using about ten times fewer parameters, and also outperforms GPT-4o, underscoring the value of targeted post-training for clinical research workflows.

LLMs for clinical trial design and optimization

We present all results for trial design tasks in Figure 3. To benchmark LLM performance on these tasks, we include open source models Qwen3-8B [30], Llama3-8B, and Llama3-70B [29], together with proprietary models GPT-4o-mini, GPT-4o, and the reasoning model o3-mini. For comparison, we also develop domain-adapted LLMs trained on `TrialPanorama` using the Qwen3-8B and Llama3-8B bases, which yield Qwen3-8B-TP and Llama3-8B-TP, where "TP" is short for `TrialPanorama`. Figures 3a and 3b illustrate the construction of the three trial design tasks. Each task presents the target trial setup on the left, including the disease area, treatment modalities, and study objective, followed by a multiple-choice question that mirrors common decisions made during protocol development. For arm design, models must determine which treatment arms best match the target trial, considering both the number of arms and the treatment composition. For eligibility criteria design, models must choose the inclusion and exclusion criteria that are most appropriate for the stated objective and patient population. For endpoint design, models must select the endpoint option that aligns with the clinical goal of the trial.

Figures 3c show the results of the three trial design tasks. Across arm design, eligibility criteria design, and endpoint design, proprietary frontier models outperform the open source baselines, while the domain-adapted models achieve the strongest overall accuracy. For arm design, the best open source model reaches 85.0%, whereas GPT-4o and O3-mini reach 86.3% and 89.0%, respectively. The domain-adapted models further improve performance, achieving 89.0% for Qwen3-8B-TP and 90.0% for Llama3-8B-TP. A similar pattern is observed in eligibility criteria design, where the baseline models plateau between 55.0% and 72.6%, while O3-mini reaches 84.4%. Domain adaptation again yields the highest accuracy, with Qwen3 8B TP at 87.1%. Endpoint design shows the largest gap between open source baselines and specialized models. Baseline models remain below 62.0%, whereas O3-mini reaches 69.1%. The domain-adapted variants achieve 74.3% and 78.2%, respectively, demonstrating consistent gains from tailoring the models to the clinical trial domain. These results indicate that domain-adapted LLMs provide the most reliable performance across diverse components of trial design.

We further present Qwen3-8B-TP-generated free-text outputs for the trial design tasks in Supplementary Tables 12, 13, and 14, illustrating the model's ability to generate clinically plausible trial design components in open-ended settings. Supplementary Note 3 provides a qualitative analysis of the evaluated cases, including the six non-aligned outputs. Although this analysis is based on a limited sample and is intended for qualitative illustration rather than quantitative evaluation, most non-aligned outputs remained clinically plausible, with only one case exhibiting a substantive design divergence.

Figure 3d shows the construction of the sample size estimation task, which requires multi-step statistical reasoning rather than pattern matching. The model must interpret a structured trial synopsis that includes the group definitions for control and experimental arms, along with the statistical assumptions needed for power analysis. Given this setup, the model must compute the required sample size that satisfies the primary power requirement. We assess model performance by computing the mean absolute error (MAE) of the model-estimated sample size against the original trial's target accrual, which was made by human statisticians. As shown in Figure 3e, open source models exhibit large estimation errors, with mean absolute error ranging from 575.4 to 1769.6. Proprietary frontier models reduce the error substantially, reaching 411.4

for GPT-4o-mini and 452.1 for GPT-4o. Domain-adapted models achieve the best accuracy overall, with a mean absolute error of 356.3 for Llama3-8B-TP and 393.6 for Qwen3-8B-TP. These results highlight both the difficulty of this reasoning-heavy task and the effectiveness of targeted domain adaptation for improving statistical modeling performance.

Figure 3f shows the trial completion assessment task, where the model predicts whether a clinical trial will terminate early and, if so, selects the most plausible reason among options such as enrollment issues, safety concerns, lack of efficacy, or operational problems. The input includes structured trial attributes such as study type, phase, allocation, intervention model, eligibility criteria, outcome measurements, and arm design, requiring the model to integrate heterogeneous signals before making a binary prediction and generating a rationalization. As shown in the left panel, open source models achieve balanced accuracy between 52.2% and 56.4%, while GPT-4o reaches 59.3%. Domain-adapted models again deliver the highest performance, with Llama3-8B-TP reaching 74.3% and Qwen3-8B-TP reaching 72.8%. The right panel reports the rationalization accuracy, which is substantially more challenging. Baseline models remain below 26.0%, whereas the domain-adapted variants reach 26.0% for Llama3-8B-TP and 31.1% for Qwen3-8B-TP. These results indicate that domain adaptation improves both the termination prediction and the ability to identify the underlying reason, even though rationalization remains a challenging task.

LLMs for systematic literature review of clinical trials

Systematic literature review (SLR) is a foundational activity in clinical research, yet it remains one of the most labor-intensive and time-consuming stages of the trial lifecycle. In early trial planning, SLR supports evidence-grounded decisions by mapping the landscape of prior studies, identifying safety signals and adverse events in relevant patient groups, and establishing realistic benchmarks for study design. In later stages, SLR enables the integration of clinical evidence across historical trials to inform guideline development and clinical practice. To capture these essential workflows, we design a set of SLR tasks centered on three key steps: study search, study screening, and evidence summarization. These tasks, illustrated in Figure 4, provide a structured benchmark for evaluating how well LLMs can assist researchers throughout the clinical evidence synthesis pipeline.

Figure 4a illustrates the setup of the study search task. Given a review topic specified through the PICO framework, including the target population, intervention, comparison group, and outcome measures, the model must generate an effective search query for a literature database such as PubMed in order to retrieve as many relevant clinical studies as possible. We evaluate retrieval performance using Recall@100 and Recall@500, which measure the coverage of ground truth studies (i.e., the ones that were included in the review paper we reference in the test case) within the top retrieved results. As shown in Figure 4b, proprietary models and domain-adapted models achieve markedly stronger coverage than general open source baselines. GPT-4o and GPT-4o mini outperform Qwen3 and Llama models at both retrieval depths, and the specialized models trained on our dataset Qwen3-8B-TP and Llama3-8B-TP provide further gains. Under Recall@100, Qwen3-8B-TP reaches approximately 32% and Llama3-8B-TP reaches approximately 31%, compared with approximately 17% for Qwen3-8B and approximately 19% for Llama3-8B. Under Recall@500, Qwen3-8B-TP attains approximately 57%, and Llama3-8B-TP attains approximately 54%, while the strongest proprietary baseline GPT-4o reaches approximately 47%. These results show that targeted post-training on our clinical trial corpus substantially enhances an LLM's ability to generate precise and comprehensive search queries for SLR workflows.

Figure 4c presents the setup of the study screening task. Given a set of papers retrieved for a review topic and the associated inclusion and exclusion criteria specified by the review, the model must decide for each paper whether it should be included or excluded. This task reflects the core filtering stage of an SLR workflow, where researchers determine which studies meet the methodological and clinical requirements of the review. We evaluate model performance using precision and recall, which together capture how accurately a model identifies eligible studies without introducing excessive false positives or false negatives. As shown in Figure 4d, proprietary models again surpass general-purpose open source LLMs, and the domain-adapted models yield the strongest overall performance. O3-mini and GPT-4o show clear gains over Qwen3 and Llama baselines, but the TP variants provide the largest improvements. Llama3-8B-TP achieves a precision of approximately 72%, and Qwen3-8B-TP reaches approximately 69%, compared with approximately 55% to 60% for the base open source models. In terms of recall, Qwen3-8B-TP attains approximately 86%, and Llama3-8B-TP reaches approximately 72%, again exceeding both open source baselines and proprietary models. These results indicate that targeted post-training substantially improves an LLM's ability to apply complex inclusion and exclusion rules and reliably identify relevant studies for SLR workflows.

Figure 4e shows the setup of the evidence summarization task. Given a review topic specified through the PICO framework and the set of studies included after screening, the model is presented with multiple candidate evidence statements and must identify which one accurately reflects the clinical finding, or determine that none of the candidates is correct. This formulation mirrors the synthesis stage in SLR workflows, where reviewers must discern which textual statements faithfully represent the aggregated evidence. We evaluate model performance using accuracy and macro F1, which jointly capture the ability to select the correct evidence statement while avoiding false selections when no option is appropriate. As shown in Figure 4f, proprietary models outperform general open source baselines, and domain-adapted models again achieve the strongest results. GPT-4o and O3-mini provide clear improvements over Qwen3 and Llama models, but the TP variants show the most substantial gains. Qwen3-8B-TP reaches an accuracy of approximately 82% and a macro F1 of approximately 81%, the highest among all evaluated models. Llama3-8B-TP also performs well with accuracy near 76% and macro F1 near 75%, compared with accuracy values between 67% and 74% for the general-purpose baselines. These results demonstrate that targeted post-training on clinical trial data significantly enhances an LLM's ability to distinguish accurate evidence from distractors, a critical capability for reliable evidence synthesis in SLR workflows.

Discussion

`TrialPanorama` drives the development of LLMs for accelerating drug development and improving population health. This paper addresses two challenges: first, developing effective LLMs requires large-scale, high-quality domain-specific data, yet such resources have historically been fragmented or unavailable. Second, robust data pipelines are needed to transform raw clinical research data into training-ready corpora for post-training LLMs, including supervised fine-tuning (SFT) and reinforcement learning with verifiable rewards (RLVR). Specifically, we start by assembling over 1.6 million clinical trial records and 30K systematic literature reviews from literature databases, ClinicalTrials.gov, and additional global trial registries. The collected data are further processed into `TrialPanorama`, which includes a knowledge graph linking trials, publications, and reviews, as well as a relational database capturing structured trial design elements.

Using this resource, we demonstrate how developers can efficiently extract structured supervision and construct development datasets for training and evaluating LLMs across five trial design and optimization tasks and three systematic review tasks. Our benchmark compares eight open and proprietary models alongside `TrialPanorama`-adapted LLMs. The results consistently show that domain-adapted models outperform generic baselines in predicting and reproducing historical clinical research decisions documented in real-world trial practice. Through the combination of `TrialPanorama` and a unified SFT plus RLVR pipeline, we observe substantial gains in reasoning quality. In particular, our 8B-level models surpass a 70B generic model and a proprietary reasoning model such as o3-mini, underscoring that task-aligned data and verifiable reward signals can be more consequential than scale alone for modeling historical clinical research decision patterns. We note that benchmark labels derived from historical trial records (e.g., sample size and trial termination outcomes) reflect decisions made in real-world clinical development settings and should be interpreted as modeling historical practice rather than necessarily representing optimal trial design choices.

This study has several limitations. First, the trial coverage of `TrialPanorama` remains uneven, particularly for regions with limited reporting, and the resulting representativity may vary across therapeutic areas and countries. Second, the data pipeline can suffer from low-quality source trials and the errors from heuristics and AI models used for data extraction and standardization. Third, we adopt proxy metrics to enable reproducible benchmarking, but these abstractions cannot capture the full complexity of clinical decision-making. Similarly, post-training approaches such as RLVR apply only to tasks with clearly verifiable labels, leaving many clinically important tasks outside the current scope. Accordingly, we emphasize that TrialPanorama-based models are intended to be used as decision support for specific steps in clinical trials with human experts in the loop, and users should remain aware of the risks of automation bias when interpreting model outputs. Finally, registry data reflect design practices at the time of registration, which may not align with evolving standards of care or contemporary regulatory expectations. Without continual updates, models trained on historical patterns may not fully capture emerging paradigms such as decentralized, adaptive, or platform trials.

In conclusion, this work demonstrates an end-to-end pipeline for developing LLMs tailored to clinical research, spanning data construction, knowledge integration, and model adaptation. By building on a large-scale, queryable database and combining supervised finetuning with reinforcement learning using verifiable rewards, we show that domain-aligned data and post-training strategies can meaningfully elevate model performance on core clinical research tasks. We envision that such data foundations, together with continued advances in algorithms, will play a central role in enabling reliable and impactful AI systems for biomedicine.

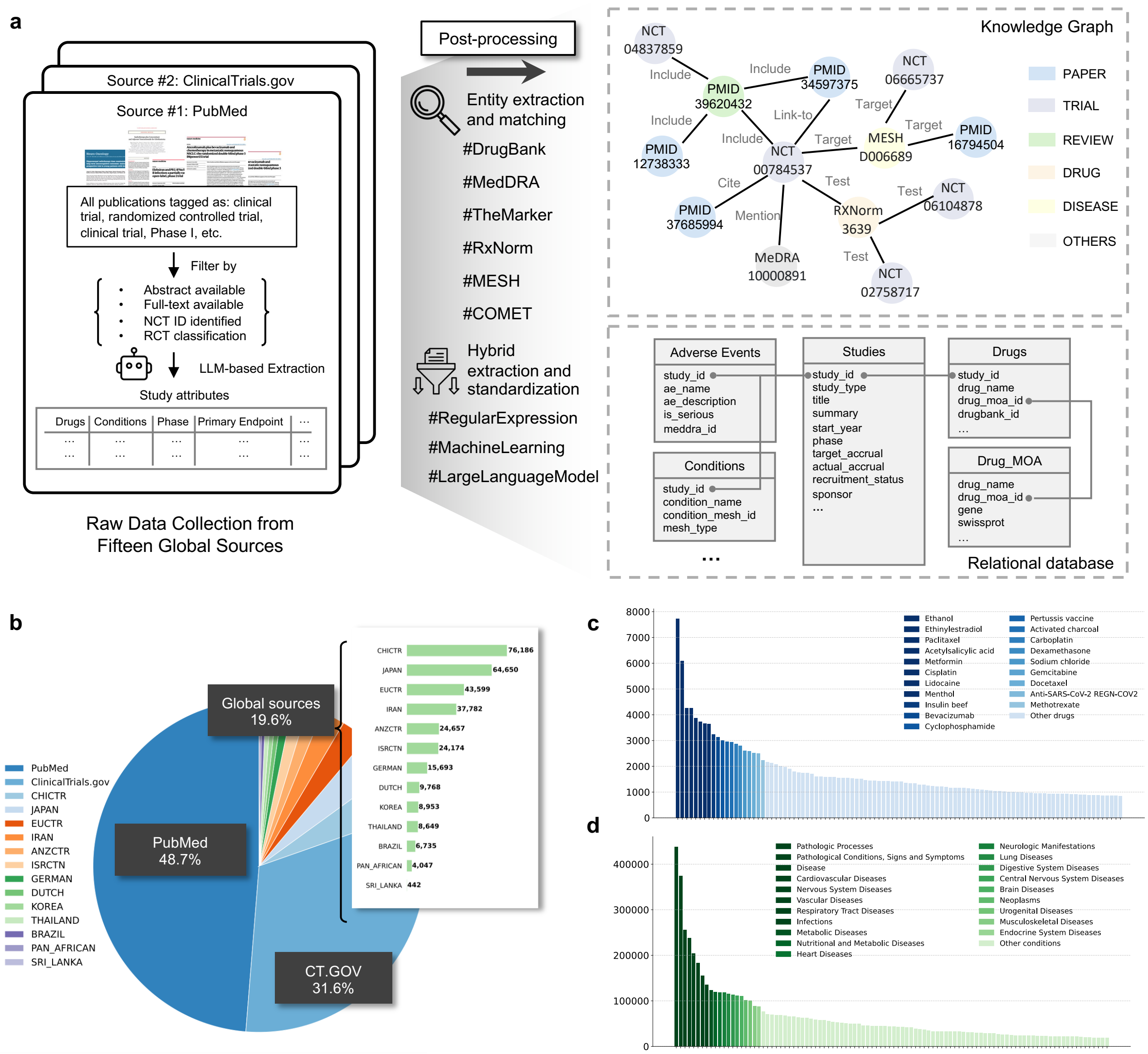

Figure 1: Overview of `TrialPanorama` construction and dataset composition. **a**, Overview of the TrialPanorama data pipeline. Raw clinical research data are collected from fifteen global sources, then processed through LLM-based extraction, rule-based standardization, and ontology-grounded normalization to derive unified study attributes. The processed data are stored in both a relational database and a heterogeneous knowledge graph linking trials, publications, drugs, diseases, and outcomes. **b**, Distribution of data sources in `TrialPanorama` showing contributions from PubMed, ClinicalTrials.gov, and aggregated global registries, together with a country-level breakdown of non-US registries. **c**, Distribution of the most frequent drug entities, illustrating the long tail coverage of therapeutic compounds. **d**, Distribution of the most frequent disease and condition entities, demonstrating broad therapeutic coverage across major disease areas.

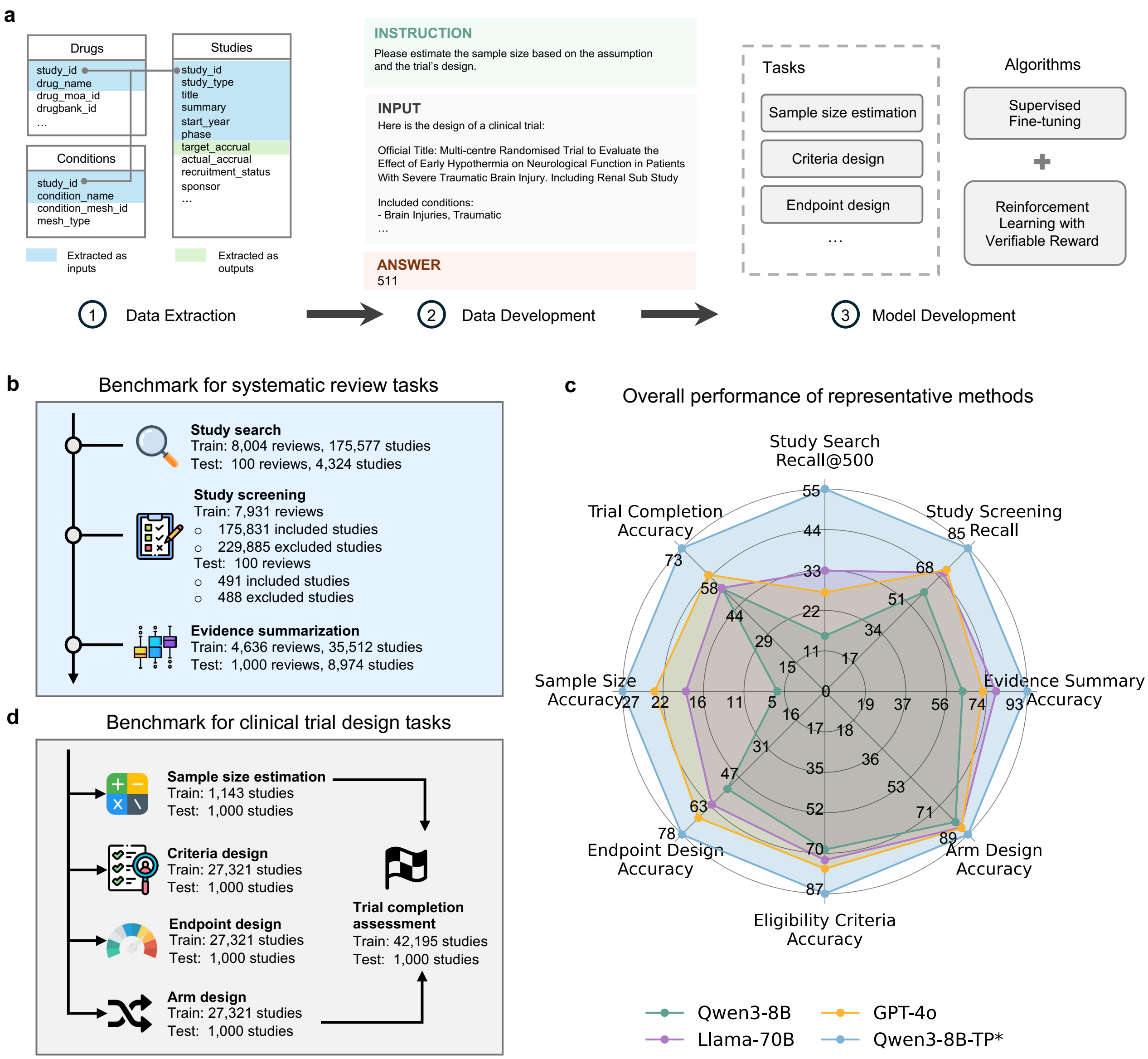

Figure 2: Benchmark construction and model performance across systematic review and clinical trial design tasks. **a**, Overview of the benchmark construction pipeline. Structured and semi-structured clinical trial records are transformed into supervised instruction examples by converting extracted study attributes into standardized input-output training pairs for tasks such as sample size estimation, eligibility criteria design, and endpoint selection. **b**, Overview of the benchmark for systematic review tasks, including study search, study screening, and evidence summarization, together with the scale of training and testing sets derived from real-world review corpora. **c**, Overall model performance across the benchmark suite. "Qwen3-8B-TP*" indicates Qwen3-8B model fine-tuned using the mixture of supervised fine-tuning (SFT) and reinforcement learning with verifiable reward (RLVR) strategies on `TrialPanorama`. Results show that domain-adapted models consistently outperform generic baselines across both systematic review and trial design tasks. **d**, Overview of benchmark tasks for clinical trial design, including sample size estimation, eligibility criteria design, endpoint selection, arm design, and trial completion assessment, along with the number of training and testing instances for each task.

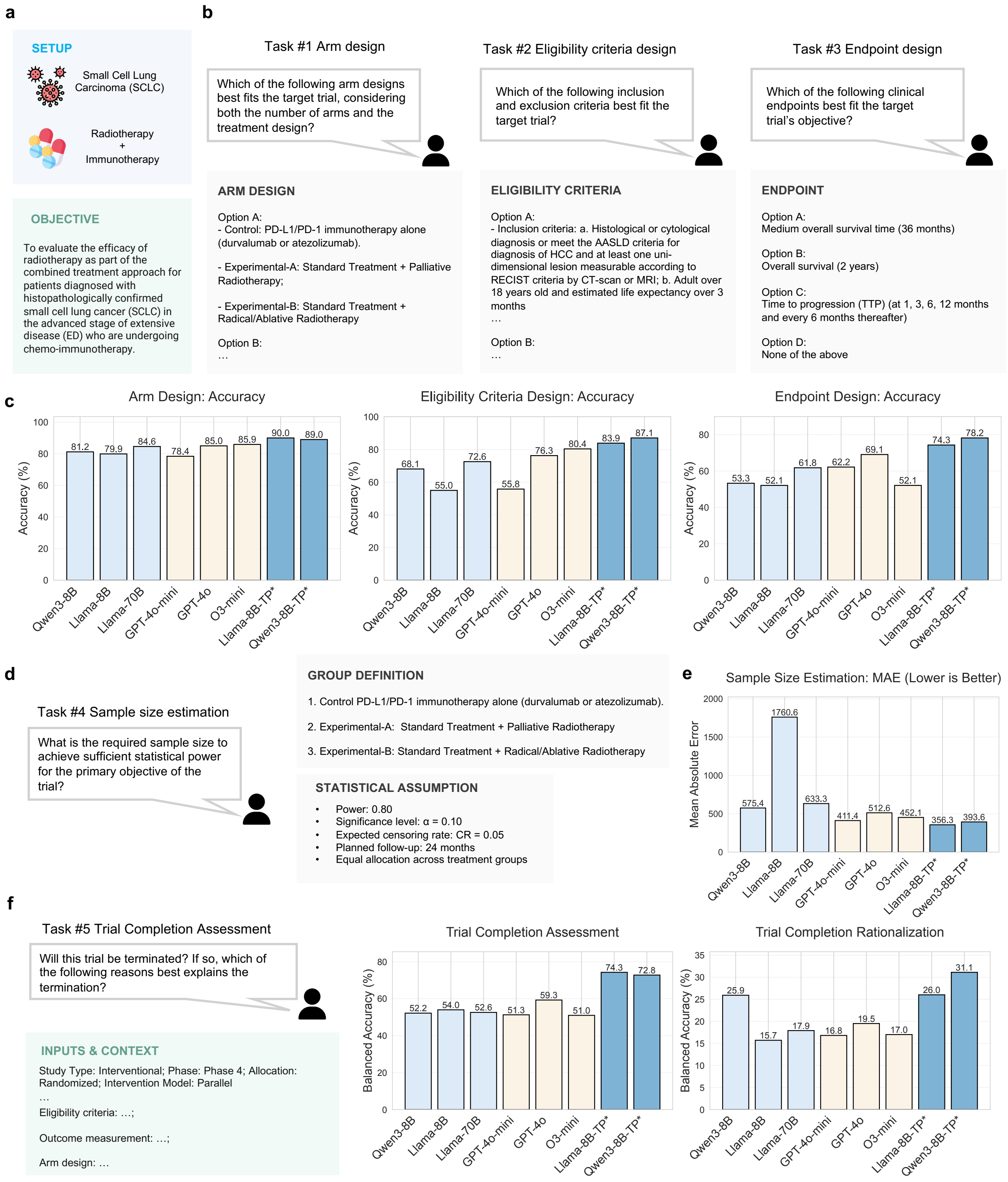

Figure 3: Benchmark tasks for clinical trial design and comparative performance of large language models. **a**, Example trial setup illustrating how key attributes, such as indication, interventions, and study objective, form the context for downstream design tasks. **b**, Illustrations of three multiple-choice design tasks: arm design, eligibility criteria design, and endpoint design, where the model must select the option that best aligns with the target trial. **c**, Accuracy of eight models on arm design, eligibility criteria design, and endpoint design, showing clear gains from domain-adapted training. **d**, Illustration of the sample size estimation task, including group definitions and statistical assumptions used to construct ground truth labels. **e**, Mean absolute error for sample size estimation across eight models, where lower error indicates better performance. **f**, Trial completion assessment task and corresponding performance, evaluated by balanced accuracy for termination prediction and rationale selection, highlighting improvements in models trained on `TrialPanorama`.

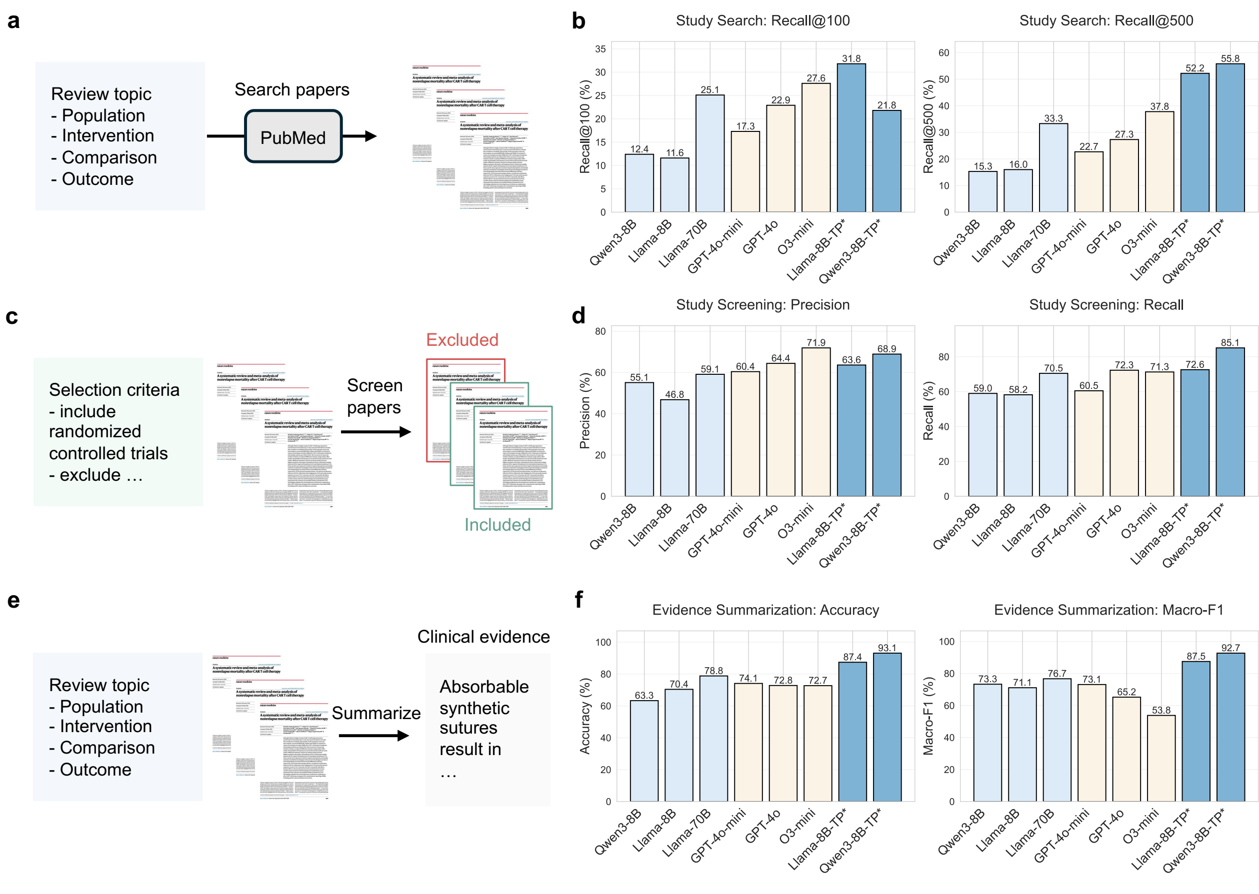

Figure 4: Systematic review task setup and model performance across study search, study screening, and evidence summarization. **a**, Study search task formulation: given a review topic specified by the PICO elements, the model must generate an effective PubMed query to retrieve relevant clinical studies. **b**, Performance of eight models on study search measured by Recall at one hundred and Recall at five hundred, highlighting substantial gains from domain-adapted training. **c**, Study screening task formulation: given review selection criteria and candidate studies, the model classifies each study as included or excluded. **d**, Precision and recall for study screening across models, showing improvements in both specificity and sensitivity with domain-adapted models. **e**, Evidence summarization task formulation: given a review topic and selected studies, the model must identify the correct evidence statement among multiple candidates or select none if no option is supported. **f**, Accuracy and Macro F1 for evidence summarization, demonstrating strong benefits of task-aligned training for fine-grained evidence reasoning.

Datasets	Sample size	Data source	Design	Result	Outcome	Relation	Ontology	Design Benchmark	Review Benchmark
TrialPanorama (Ours)	1.6M	Multiple (15 sources)	✓	✓	✓	✓	✓	✓	✓
TOP	10K	ClinicalTrials.gov	✗	✗	✓	✗	✗	✓	✗
CTKG	8K	ClinicalTrials.gov	✓	✗	✗	✗	✓	✗	✗
ClinicalTrials.gov	56K	ClinicalTrials.gov	✗	✓	✓	✗	✗	✗	✗
CTO	125K	ClinicalTrials.gov	✗	✓	✓	✗	✗	✗	✗
TrialBench	400K	ClinicalTrials.gov	✓	✗	✓	✗	✓	✓	✗
CARE	700 pubs	PubMed	✗	✓	✓	✗	✗	✗	✗
PubMedKG	480K	PubMed, ClinicalTrials.gov	✗	✗	✗	✓	✗	✗	✗
TrialReviewBench	100 reviews	PubMed, ClinicalTrials.gov	✗	✗	✗	✓	✗	✗	✓
LEADSInstruct	21K reviews	PubMed, ClinicalTrials.gov	✗	✗	✗	✓	✗	✗	✓

Table 1: Comparison of `TrialPanorama` with prior clinical trial datasets and benchmarks. The table summarizes sample size, data sources, and coverage of key structured components, including design attributes, trial results, clinical outcomes, study publication relations, and ontology linkage. It also highlights whether each resource provides benchmark tasks for trial design or systematic review workflows, showing that `TrialPanorama` offers the most comprehensive coverage across all dimensions.

Method

Data collection

To construct our benchmark dataset, we crawled clinical trial and publication records from a diverse set of publicly accessible registries and databases. The data URLs and the date of the most recent crawl are listed in Supplementary Table 1. After collection, the raw data were subjected to systematic cleaning procedures to ensure consistency and reliability across sources: (1) Intra-source de-duplication: Duplicate entries within the same registry were detected and removed using unique identifiers (such as trial registration numbers), and corroborated using additional metadata like titles. (2) Inter-source de-duplication: Clinical trials registered in multiple registries were identified via fuzzy matching on titles. Duplicate records across sources were merged into a single one. (3) Field normalization: Metadata fields such as trial status, registration date, and condition labels were standardized across sources to ensure consistent querying. (4) Missing value handling: Entries with missing or incomplete critical fields were flagged. These operations minimized noise, reduced redundancy, and created a unified and clean dataset suitable for benchmarking tasks.

Database building

The overall database schema is illustrated in Supplementary Figure 1. `TrialPanorama` is organized into a set of structured tables that together provide a unified view of global clinical research data. The Studies table serves as the central entry for each trial and aggregates identifiers from registries and publications, trial design attributes such as phase and sponsor, and primary text fields including titles and summaries. Several complementary tables enrich each study with standardized clinical entities. The Conditions table stores disease terms mapped to Medical Subject Headings, including both primary concepts and hierarchical ancestors. The Drugs table provides a normalized representation of pharmacologic interventions, including cleaned drug names, RxNorm and DrugBank identifiers, and approval status from major regulatory agencies. The pipeline and mapping statistics for drug name normalization are in Supplementary Figure 2, and Supplementary Table 2, respectively. The Disposition table captures the arm structure of each trial by pairing treatment groups with intervention descriptions, while the Adverse Events table records arm-specific safety findings from ClinicalTrials.gov, including frequencies, seriousness categories, and MedDRA codes.

To support evidence extraction and comparative analysis, we constructed additional tables centered on trial findings and clinical endpoints. The Trial Results table provides PICO structured summaries of outcome data at the arm level, combining extractions from PubMed abstracts with structured results from ClinicalTrials.gov. The Trial Outcomes table assigns unified outcome types to completed and terminated studies, including positive and negative classifications and fine-grained termination reasons. The Endpoints table standardizes primary outcomes by mapping them to COMET domains and subdomains, creating a consistent representation of clinical endpoints across heterogeneous sources. The Biomarkers table links text spans from trial descriptions to curated therapeutic biomarkers from TheMarker and records biomarker name, functional class, biomarker type, and associated genes when applicable. The distribution of the most frequent biomarkers is shown in Supplementary Figure 3.

Finally, the Relations table encodes connections among systematic reviews, cited studies, and registry records. Each entry is a structured triple that captures include, exclude, awaiting assessment, or cross registry link relationships. Together, these tables form a comprehensive relational database that supports large-scale analysis of trial design, evidence synthesis, and clinical development patterns. Detailed procedures for extraction, normalization, and validation are reported in Supplementary Section 1. The statistics of the relation types are in Supplementary Table 3.

Trial design tasks: arm design

This task assesses whether the model can identify the correct treatment arms and intervention details for a given clinical trial. Based on the trial's title and brief summary, the model must choose among four candidate descriptions the one that accurately represents the trial's arm labels, arm types, and assigned interventions. An example instance is shown in Supplementary Table 4. Given this setup, the model is provided the official trial title and summary, followed by four arm descriptions that list the group labels, intervention types, and associated treatments. It must return the label corresponding to the correct description.

To construct the dataset, we start with sets of up to four trial identifiers drawn from trials cited together in the same Cochrane review. For each identifier, a script extracts the arm groups and individual intervention details from our trial metadata. If fewer than four candidates are available, additional trials are added at random (with a fixed seed for reproducibility). To create challenging distractors, incorrect descriptions are drawn from the same arm and intervention information of other trials cited by that review. All items are then sorted by the numeric portion of the trial identifier and partitioned so that the latest 1,000 trials form the test set, the next 500 form validation, and the rest form training. We report accuracy and macro-F1 to account for class imbalance across answer choices.

Trial design tasks: eligibility criteria design

This task measures the model's ability to match a trial's inclusion and exclusion criteria. Given a trial's title and summary, the model must select, from four text options, the one that accurately lists the trial's eligibility rules. An example instance is shown in Supplementary Table 5. In this task, the model receives the trial title and summary along with four candidate eligibility criterion sets. It must output the correct option label.

To construct the dataset, we follow the same approach as in Arm Design: we form candidate sets from trials cited by the same Cochrane review, extract each trial's eligibility rules from our metadata, pad with random trials when needed, and draw distractors from other cited trials to ensure relevance. Examples are then sorted by the numeric part of the trial identifier and split into test (latest 1,000), validation (next 500), and training (remainder). We report Accuracy and macro-F1 to account for class imbalance across answer choices.

Trial design tasks: endpoint design

This task evaluates whether the model can identify the primary outcome measures of a clinical trial. Based on the trial title and summary, the model must pick the option that lists the correct primary endpoints. An example instance is shown in Supplementary Table 6. The input consists of the trial title and summary plus four outcome descriptions that concatenate the primary measures and any associated timeframes. The model must choose the correct label.

The data construction steps are: for trials cited together provide candidate sets, we extract each trial's primary outcomes from our metadata, pad with random examples if needed, and sample distractor options from other cited trials. The resulting items are ordered by the numeric portion of the trial identifier and divided into test (latest 1,000), validation (next 500), and training splits. We report Accuracy and macro-F1 to account for class imbalance across answer choices.

Trial design tasks: sample size estimation

This task evaluates the model's quantitative reasoning by asking it to estimate the number of participants required for a clinical trial based on its design and stated statistical assumptions. Success requires understanding trial objectives, translating textual assumptions (effect size, significance level, power, dropout) into a numerical enrollment target, and applying standard sample-size formulas. An example instance is shown in Supplementary Table 7. As input, the model receives: a narrative description of the trial design: official title, conditions, arm groups, interventions and primary outcomes, as well as a concise summary of the statistical assumptions extracted from the protocol text (treatment effect, alpha, power, expected dropout). The model must output a single integer corresponding to the estimated sample size.

To construction the dataset, we linked 5,737 PubMed protocols to their NCTIDs and identified 4,315 full-text sections titled "sample size" together with 5,678 CT.gov records containing enrollment counts. Matching the reported CT.gov enrollment with the corresponding paper section yielded 2 143 examples where the protocol paragraph explicitly mentions the official enrollment number. For each example, we used a deterministic prompt to summarize the paragraph into (1) effect-size assumptions, (2) statistical parameters (significance level, power, dropout rate), and (3) the true enrollment. We then generated a question by combining trial metadata with these summarized assumptions. Finally, all examples were sorted by the numeric part of the NCTID; the most recent 1,000 form the test set, the next 500 the validation set, and the remainder the training set. In evaluation, we report (1) the mean absolute error between predicted and actual enrollments, and (2) the proportion of trials for which the model's estimate lies within ±20 % of the true sample size.

Trial design tasks: trial completion assessment

This task simulates a real-world feasibility analysis during clinical trial planning, where the model must assess whether a trial is likely to reach completion or be terminated early based on its protocol design. In addition, if a trial is predicted to terminate, the model is asked to identify the most likely reason for termination from a predefined set of categories (e.g., enrollment issues, safety concerns, lack of efficacy, operational problems). This reflects practical needs in the early design stage when sponsors assess trial viability. An example instance is shown in Supplementary Table 8. Each example consists of structured metadata and protocol text for a registered interventional study, including trial phase, number of arms, masking, intervention model, primary purpose, arm descriptions, and detailed eligibility criteria. The output is a binary label indicating whether the trial was completed or terminated. For trials labeled as `terminated`, the model must also predict the most likely termination reason from one of several standardized categories: lack of efficacy, enrollment issues, safety issues, feasibility issues, and regulatory issues.

We construct the dataset using completed and terminated interventional trials from ClinicalTrials.gov with known outcomes. Structured fields are extracted from the studies, disposition, eligibility criteria, and arms tables in our trial database. Only trials with definitive outcome labels (i.e., Completed or Terminated) and non-missing eligibility criteria are retained. Termination reasons are normalized into broad categories using a curated taxonomy. To avoid label leakage, we exclude any outcomes or termination reasons from the input text fields. The extraction workflow is illustrated in Supplementary Figure 4. Finally, we evaluate trial completion prediction using accuracy. For terminated trials, termination reason classification is evaluated using macro-F1 to balance across categories with different frequencies.

Systematic literature review tasks: study search

This task simulates the study search step in conducting systematic reviews, where the model must comprehend a given research question and synthesize a corresponding search query to retrieve relevant studies for inclusion. The key motivation behind this task is to streamline a traditionally labor-intensive process that typically requires the expertise of trained librarians, particularly due to the complexity and variability of search query formats across literature databases. Automating this step can significantly accelerate systematic review workflows. An example instance is presented in Supplementary Table 9.

The input for this task includes the setup of a systematic review, consisting of the research background, objective, and study selection criteria. The model is expected to generate a structured search query, which is then submitted to the PubMed Search API to retrieve the top 500 results and their associated PubMed IDs. A successful output should go beyond simple keyword extraction, incorporating semantic understanding and augmentation of the input to produce a comprehensive and effective query.

We constructed the dataset using a relation table that links systematic reviews to primary studies. Specifically, we focused on instances with the relation type `include`, indicating that the study was selected for inclusion in the review. For each review, we parsed its full text to extract relevant sections and used the linked PubMed abstracts as ground-truth targets. This process resulted in a dataset of 8,104 examples derived from systematic review articles, covering more than 180,000 included PubMed studies. We reserved the most recent 100 reviews as a held-out test set.

We evaluate this task as a retrieval problem by measuring how many ground-truth studies appear in the top search results produced by the model-generated query. Specifically, we report Recall@100 and Recall@500, reflecting the number of included studies retrieved among the top 100 and 500 PubMed results, respectively.

Systematic literature review tasks: study screening

This task simulates the study selection step in a systematic review pipeline, where the model must assess the eligibility of candidate studies for inclusion based on the review's predefined criteria. Given a systematic review's background, objective, and eligibility criteria, along with a set of candidate studies retrieved from PubMed (each with a title and structured abstract), the model determines whether each study should be `included` or `excluded`. This step is often labor-intensive and requires detailed reasoning across clinical nuances and protocol designs. An example instance is shown in Supplementary Table 10.

The input to this task consists of the systematic review protocol (comprising the review's background, objective, and selection criteria) and a list of candidate studies, each represented by its title and structured abstract. The output is a binary label (`included` or `excluded`) for each candidate study, indicating whether it meets the review's eligibility criteria. Accurate performance requires the model to match eligibility conditions against sometimes subtle or incomplete cues in the candidate study abstracts.

We construct the dataset using reviews that have explicit inclusion and exclusion annotations for candidate studies, available from the `relations` table in our clinical trial knowledge base. For each review, we extract its background, objective, and selection criteria from the abstract, and retrieve associated candidate studies along with their labels (`included` or `excluded`) based on the relation types `include` and `exclude`. To ensure a balanced and informative evaluation, we downsample the candidate pool for each review to include a random mix of included and excluded studies, preserving diversity while controlling for class imbalance. Each instance contains up to 10 candidate studies, each with a title and structured abstract, paired with a single review protocol.

We evaluate model performance using three metrics: accuracy, precision, and recall. Accuracy reflects the overall proportion of correctly classified studies. Precision measures the proportion of correctly identified `included` studies among all studies predicted as included, while recall measures the proportion of truly `included` studies that were correctly identified by the model. These metrics jointly assess both correctness and completeness of model screening decisions.

Systematic literature review tasks: evidence summarization

This task simulates the final synthesis step of a systematic review, in which the model must draw a conclusion from the included studies' findings. Given a self-contained multiple-choice question based on a Cochrane review's objectives, the model selects the answer option that aligns with the review's main conclusion. An example instance is shown in Supplementary Table 11. The input to the evidence summarization task consists of a self-contained multiple-choice question generated from a Cochrane review abstract (including its answer options A, B, C, ...) together with the titles and full structured abstracts of all studies included in that review (ranging from 1 to 20 PMIDs). The model's output is a single label (A, B, C, ...) corresponding to the correct conclusion. Successfully performing this task requires the model to comprehend scientific text, aggregate evidence across multiple documents, and map study findings to the correct conclusion.

The dataset construction steps are: First, Cochrane review abstracts are processed by GPT-4o using a prompt that instructs the model to generate a self-contained multiple-choice question based on the review objectives, to produce answer options covering plausible outcomes, and to output only valid JSON with the keys `question`, `options`, and `answer`. The prompt also forbids referencing the original review by phrases such as "based on" to ensure questions are fully self-contained, and the model is run at temperature 0 to guarantee deterministic outputs. Next, for each generated question, we retrieve the included-study PMIDs from the review metadata and load the corresponding titles and structured abstracts from our PubMed cache, discarding any entries with zero or more than twenty included studies. Finally, the remaining N examples are sorted by numeric PMID (serving as a proxy for publication date); the most recent 1,000 examples form the *test* set, the preceding 500 form the *validation* set, and the remainder form the *training* set. In evaluation, we report accuracy and macro-F1 to account for class imbalance across answer choices.

Model development

In experiments, we apply a two-stage post-training pipeline combining supervised fine-tuning (SFT) and reinforcement learning with verifiable rewards (RLVR), where SFT injects domain knowledge into LLMs and RLVR further refines task-specific reasoning. We instantiate this pipeline using two representative base models: LLaMA-3-8B-Instruct [29] and Qwen3-8B [30].

Supervised Fine-tuning (SFT)

Supervised fine-tuning was performed in two sequential phases. In the first phase, the model was trained only on the tasks with high-volume datasets, specifically trial design and trial completion assessment. In the second phase, the model was trained on the remaining tasks, including sample size estimation, study search, study screening, and evidence summarization. Separating these phases ensured that high-volume tasks did not dominate the gradient signals, otherwise the substantially smaller tasks would struggle to be adequately learned by the model.

The model was optimized using the next token prediction objective applied only to answer tokens. Let x denote the input sequence, which includes both the instruction and the question, and let $y = (y_1, \ldots, y_K)$ denote the target answer sequence. The SFT objective is

$$\mathcal{L}_{\mathrm{SFT}} = -\sum_{t=1}^{K} \log p_\theta\big(y_t \mid y_{<t}, x\big), \tag{1}$$

so that the input x only serves as conditioning context, while the loss is computed exclusively over the answer tokens y.

We implement the SFT recipe by using the LLaMA Factory framework [36]. Training used a context length of 8192 tokens and a peak learning rate of 1×10^{-6} under a cosine decay schedule. The AdamW optimizer was employed with a batch size of 16 and numerical precision was set to bfloat16. To increase computational efficiency, DeepSpeed ZeRO-3 [37] and FlashAttention-2 [38] were incorporated. Phase 1 was trained for three epochs, and Phase 2 continued for an additional ten epochs.

Reinforcement Learning with Verifiable Rewards (RLVR)

After SFT, we further apply RLVR to optimize the models for sample size estimation and study search tasks, where SFT alone is insufficient to achieve strong performance. At a high level, SFT teaches the model to imitate reference outputs, whereas RLVR enables the model to improve through action-based optimization, providing stronger supervision for tasks requiring deep reasoning and iterative refinement. For example, study search requires generating effective search queries, and RLVR allows the model to interact with the publication search engine during training and learn strategies for formulating high-quality queries through feedback. In RLVR, the policy model is the LLM itself, which takes the current state as input and outputs an action, while the environment evaluates the action and returns reward feedback. Through iterative interaction with the environment, the policy model learns to improve its decisions over time. In our setting, the key challenge is to formulate these components appropriately for each task to enable effective training. We describe the RLVR formulation for study search and sample size estimation below.

For study search, the environment is the PubMed search engine, which takes structured search queries generated by the LLM and returns search results used for reward computation. Given the included studies from the reference review paper as ground truth, we calculate Recall@100 based on the proportion of ground-truth studies retrieved within the top 100 search results, and use this as the reward signal. For sample size estimation, the action is the sample size predicted by the model, and the reward is computed based on its deviation from the reference trial sample size, which was originally determined by biostatisticians in historical studies. In addition, we apply a format reward to both tasks. Specifically, the model is required to present its final answer within an <answer> ... </answer> tag pair. Outputs that fail to satisfy this requirement receive a penalty of -2, whereas correctly formatted outputs receive a reward of 0.1.

In our experiments, we adopt Group Relative Policy Optimization (GRPO) as the learning algorithm for RLVR [39], implemented using the VeRL framework [40]. Let x denote an input prompt sampled from the training set, and let π_θ represent the policy model parameterized by θ. For each input x, the policy generates a group of G rollout sequences $\{y_i\}_{i=1}^{G}$ by sampling from the previous policy $\pi_{\theta_{\text{old}}}$, where each rollout $y_i = (y_{i,1}, y_{i,2}, \ldots, y_{i,T_i})$ is a complete output token sequence with length T_i. Each rollout is then evaluated by the task-specific reward function to produce a scalar reward r_i.

GRPO optimizes the policy by comparing the relative quality of rollouts within each sampled group. Specifically, rewards are normalized across the G rollouts to compute a group-relative advantage score for each sample:

$$A_i = \frac{r_i - \text{mean}(\{r_1, \ldots, r_G\})}{\text{std}(\{r_1, \ldots, r_G\})}. \tag{2}$$

The policy is then updated by maximizing the clipped surrogate objective:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{x,\ \{y_i\}_{i=1}^{G} \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\tfrac{1}{G} \textstyle\sum_{i=1}^{G} \tfrac{1}{T_i} \sum_{t=1}^{T_i} \min\big(\rho_{i,t} A_i,\ \text{clip}(\rho_{i,t}, 1-\varepsilon, 1+\varepsilon) A_i\big) \right] \tag{3}$$

where $\rho_{i,t}$ is the token-level policy ratio defined as

$$\rho_{i,t} = \frac{\pi_\theta(y_{i,t} \mid x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t} \mid x, y_{i,<t})}. \tag{4}$$

Intuitively, GRPO encourages the model to increase the probability of rollouts that perform better relative to others in the same group, while suppressing poorer-performing rollouts, thereby improving policy quality through group-wise comparative optimization.

The hyperparameters used for RLVR training are as follows. We set the maximum prompt length to 2,048 tokens and the maximum response length to 512 tokens. Training and validation batch sizes were set to 16 and 32, respectively, with a learning rate of 1×10^{-6}. During rollout generation, we used a sampling temperature of 0.6 and nucleus sampling with top-p set to 0.95. For each input prompt, $G = 8$ rollouts were sampled for GRPO optimization. Each RLVR training stage was conducted for five epochs.

Data Availability

The database is available at `https://huggingface.co/datasets/TrialPanorama/TrialPanorama-database`. The built training and evaluation datasets are available at `https://huggingface.co/datasets/TrialPanorama/TrialPanorama-benchmark`.

Code Availability

The code for experiments used in this study is available at `https://github.com/Keiji-AI/TrialPanorama`.

Author Contributions

Z.W. and J.L. conceived the study and led the writing of the manuscript. Z.W. conducted the primary experiments, with contributions from J.L. and Q.J. Z.W., J.L., Q.J., J.G., J.P., and P.J. curated and prepared the materials and data used in the study. Z.L. and J.S. provided senior guidance, domain expertise, and critical feedback throughout the project. All authors reviewed and approved the final manuscript.

Competing Interests

The authors declare no competing financial or non-financial interests.

Acknowledgements

J.S. was partially supported by National Science Foundation (NSF) awards SCH-2205289, SCH-2014438 and IIS-2034479. Q.J. and Z.L. were partly supported by the Intramural Research Program of the NIH. The contributions of the NIH authors are considered Works of the United States Government. The findings and conclusions presented in this paper are those of the authors and do not necessarily reflect the views of the NIH or the U.S. Department of Health and Human Services.

References

[1] Martin, L., Hutchens, M., Hawkins, C. & Radnov, A. How much do clinical trials cost? *Nature Reviews Drug Discovery* **16**, 381–382 (2017).

[2] Field, A. P. & Gillett, R. How to do a meta-analysis. *British Journal of Mathematical and Statistical Psychology* **63**, 665–694 (2010).

[3] Concato, J., Shah, N. & Horwitz, R. I. Randomized, controlled trials, observational studies, and the hierarchy of research designs. In *Research Ethics*, 207–212 (Routledge, 2017).

[4] Jin, Q., Tan, C., Chen, M., Liu, X. & Huang, S. Predicting clinical trial results by implicit evidence integration. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 1461–1477 (2020).

[5] Fu, T., Huang, K., Xiao, C., Glass, L. M. & Sun, J. Hint: Hierarchical interaction network for clinical-trial-outcome predictions. *Patterns* **3** (2022).

[6] Wang, Z., Xiao, C. & Sun, J. Spot: Sequential predictive modeling of clinical trial outcome with meta-learning. In *Proceedings of the 14th ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics*, 1–11 (2023).

[7] Jin, Q. *et al.* Matching patients to clinical trials with large language models. *Nature Communications* **15**, 9074 (2024).

[8] Wornow, M. *et al.* Zero-shot clinical trial patient matching with llms. *NEJM AI* **2**, AIcs2400360 (2025).

[9] Yuan, C. *et al.* Criteria2Query: a natural language interface to clinical databases for cohort definition. *Journal of the American Medical Informatics Association* **26**, 294–305 (2019).

[10] Park, J. *et al.* Criteria2Query 3.0: Leveraging generative large language models for clinical trial eligibility query generation. *Journal of Biomedical Informatics* **154**, 104649 (2024).

[11] Wang, Z., Xiao, C. & Sun, J. Autotrial: Prompting language models for clinical trial design. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 12461–12472 (2023).

[12] Wang, Z. *et al.* Compliance and factuality of large language models for clinical research document generation. *Journal of the American Medical Informatics Association* ocaf174 (2025). URL `https://doi.org/10.1093/jamia/ocaf174`.

[13] Wang, Z. *et al.* A foundation model for human-ai collaboration in medical literature mining. *Nature Communications* **16**, 8361 (2025).

[14] Wang, Z. *et al.* Accelerating clinical evidence synthesis with large language models. *npj Digital Medicine* **8**, 509 (2025).

[15] Clinicaltrials.gov is a place to learn about clinical studies from around the world. `https://clinicaltrials.gov/`. Accessed: 2025-05-01.

[16] International clinical trials registry platform (ictrp). `https://www.who.int/tools/clinical-trials-registry-platform`. Accessed: 2025-05-01.

[17] Eu clinical trials register. `https://www.clinicaltrialsregister.eu/about.html`. Accessed: 2025-05-01.

[18] Australian new zealand clinical trials registry. `https://www.anzctr.org.au/`. Accessed: 2025-05-01.

[19] Chinese clinical trial registry. `https://www.chictr.org.cn/indexEN.html`. Accessed: 2025-05-01.

[20] Shi, X. & Du, J. Constructing a finer-grained representation of clinical trial results from clinicaltrials. gov. *Scientific Data* **11**, 41 (2024).

[21] Naik, A., Kuehl, B., Bransom, E., Downey, D. & Hope, T. Care: Extracting experimental findings from clinical literature. In *Findings of the Association for Computational Linguistics: NAACL 2024*, 4580–4596 (2024).

[22] Chen, Z. *et al.* A knowledge graph of clinical trials (ctkg). *Scientific reports* **12**, 4724 (2022).

[23] Brbić, M., Yasunaga, M., Agarwal, P. & Leskovec, J. Predicting drug outcome of population via clinical knowledge graph. *medRxiv* (2024).

[24] Xu, J. *et al.* Building a pubmed knowledge graph. *Scientific data* **7**, 205 (2020).

[25] Gao, C., Pradeepkumar, J., Das, T., Thati, S. & Sun, J. Automatically labeling $200 b life-saving datasets: A large clinical trial outcome benchmark. *arXiv preprint arXiv:2406.10292* (2024).

[26] Dunn, A. G. & Bourgeois, F. T. Is it time for computable evidence synthesis? *Journal of the American Medical Informatics Association* **27**, 972–975 (2020).

[27] Holzhauer, B. & Adewuyi, E. T. Super-covariates: Using predicted control group outcome as a covariate in randomized clinical trials. *Pharmaceutical Statistics* **22**, 1062–1075 (2023).

[28] Kim, E. S. *et al.* Broadening eligibility criteria to make clinical trials more representative: American society of clinical oncology and friends of cancer research joint research statement. *Journal of Clinical Oncology* **35**, 3737–3744 (2017).

[29] Dubey, A. *et al.* The llama 3 herd of models. *arXiv e-prints* arXiv–2407 (2024).

[30] Yang, A. *et al.* Qwen3 technical report. *arXiv preprint arXiv:2505.09388* (2025).

[31] The new drugbank: Powered by ai. `https://go.drugbank.com/`. Accessed: 2025-12-09.

[32] Meddra: Medical dictionary for regulatory activities. `https://www.meddra.org/`. Accessed: 2025-12-09.

[33] Unified medical language system (umls). `https://www.nlm.nih.gov/research/umls/rxnorm/index.html`. Accessed: 2025-12-09.

[34] Welcome to medical subject headings. `https://www.nlm.nih.gov/mesh/meshhome.html`. Accessed: 2025-12-09.

[35] Kirkham, J. J. & Williamson, P. Core outcome sets in medical research. *BMJ medicine* **1**, e000284 (2022).

[36] Zheng, Y., Zhang, R., Zhang, J., YeYanhan, Y. & Luo, Z. Llamafactory: Unified efficient fine-tuning of 100+ language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, 400–410 (2024).

[37] Rajbhandari, S., Rasley, J., Ruwase, O. & He, Y. Zero: Memory optimizations toward training trillion parameter models. In *SC20: International Conference for High Performance Computing, Networking, Storage and Analysis*, 1–16 (IEEE, 2020).

[38] Dao, T. Flashattention-2: Faster attention with better parallelism and work partitioning. In *The Twelfth International Conference on Learning Representations* (2024). URL `https://openreview.net/forum?id=mZn2Xyh9Ec`.

[39] Shao, Z. *et al.* Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300* (2024).

[40] Sheng, G. *et al.* Hybridflow: A flexible and efficient rlhf framework. In *Proceedings of the Twentieth European Conference on Computer Systems*, 1279–1297 (2025).